THE UNIVERSITY OF MELBOURNE

Student number

Semester 2 Assessment, 2024

School of Mathematics and Statistics

MAST20026 Real Analysis

Reading time: 15 minutes — Writing time: 3 hours

This exam consists of 29 pages (including this page) with 10 questions and 100 total marks

Permitted Materials

- No books, notes or other printed or handwritten material are permitted.
- No calculators are permitted.
- Any mobile phones or internet-enabled devices brought into the exam room must be **turned off** and placed on the floor under your table.

Instructions to Students

- Write your answers in the boxes provided on the exam. There is extra space you can use for answers to any question commencing on page 26. If you still do not have enough space, tick the box near the bottom of page 29 and request extra pages from an invigilator—include the question number at the top of each additional page.
- You must NOT remove this question paper, or any extra pages provided to you, at the conclusion of the examination.

Instructions to Invigilators

- Students are to write their answers on the paper. They may request extra pages if they run out of space on the exam paper.
- This exam paper contains examinable material and must be collected, together with extra pages if any, at the conclusion of the examination.

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Question 1 (8 marks)

Let p and q be primitive statements.

(a) Write a truth table for the compound statement

$$(\sim p) \iff (p \vee q).$$

(b) If $(\sim p) \iff (p \vee q)$ is true, what can you conclude about the truth value of p? Justify your answer in terms of specific lines in truth table from Part (a).

(c) Using a truth table, show that $(\sim p) \iff (p \vee q)$ is logically equivalent to $\sim (q \implies p)$.

Question 2 (9 marks)

Find the largest positive integer d such that for every $n \in \mathbb{N}$, $5^n - 8n - 1$ is divisible by d. Use induction to prove your assertion is true.

(Hint: Start by writing out $5^n - 8n - 1$ for $n = 0, 1, 2, 3$ and making a reasonable guess for the value of d.)

Question 3 (7 marks)

Listed below are the Axioms of the Real Numbers.

O1. $\forall x, y \in \mathbb{R}$ exactly one of the following is true: $x < y$, $y < x$, or $x = y$.
O2. $\forall x, y, z \in \mathbb{R}$ if $x < y$ and $y < z$ then $x < z$.
A1. $\forall x, y \in \mathbb{R} \quad x + y \in \mathbb{R}$
A2. $\forall x, y \in \mathbb{R} \quad x + y = y + x$
A3. $\forall x, y, z \in \mathbb{R} \quad (x + y) + z = x + (y + z)$
A4. $\exists 0 \in \mathbb{R} \quad \forall x \in \mathbb{R} \quad x + 0 = x$
A5. $\forall x \in \mathbb{R} \quad \exists -x \in \mathbb{R} \quad x + (-x) = 0$
M1. $\forall x, y \in \mathbb{R} \quad xy \in \mathbb{R}$
M2. $\forall x, y \in \mathbb{R} \quad xy = yx$
M3. $\forall x, y, z \in \mathbb{R} \quad (xy)z = x(yz)$
M4. $\exists 1 \in \mathbb{R} \setminus \{0\} \quad \forall x \in \mathbb{R} \quad 1x = x$
M5. $\forall x \in \mathbb{R} \setminus \{0\} \quad \exists 1/x \in \mathbb{R} \setminus \{0\} \quad x \cdot 1/x = 1$
D. $\forall x, y, z \in \mathbb{R} \quad x(y + z) = xy + xz$
OA. $\forall x, y, z \in \mathbb{R}$ if $x < y$ then $x + z < y + z$
OM. $\forall x, y \in \mathbb{R}$ if $x > 0$ and $y > 0$ then $xy > 0$
C. Every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Using only the Axioms of the Real Numbers (and rules of logic), write a formal proof showing that for all non-zero $a \in \mathbb{R}$

$$1/(1/a) = a.$$

Note. *Here and for the remainder of the exam, you may write informal proofs, and you may assume that algebra and inequalities with the real numbers behave the way you expect.*

Question 4 (9 marks)

For each of the following situations, provide a specific example for which the situation is true, and briefly justify your answer.

(a) An unbounded set E such that the set of all limit points of E is $[0, 1]$.

(b) A non-constant sequence (f_n) which is contractive and converges to -1.

(c) A function $g : \mathbb{R} \to \mathbb{R}$ such that $|g|$ is continuous on $\mathbb{R}$ but g is not continuous at a for all $a \in \mathbb{R}$.

Question 5 (12 marks)

Let $A, B \subset \mathbb{R}$ be sets which are both bounded above in $\mathbb{R}$. Let α be the least upper bound of A and β be the least upper bound of B.

(a) Show that $\max\{\alpha, \beta\}$ is an upper bound of the set $A \cup B$.

(b) Show that $\sup(A \cup B) = \max\{\alpha, \beta\}$.

(c) Is it always true that $\max\{\alpha, \beta\}$ is a limit point of $A \cup B$? Justify your answer with a proof or counter-example.

Question 6 (10 marks)

Consider the sequence (f_n) defined by

$$f_n = \sqrt{2n-1}$$

for each $n \in \mathbb{N}^+$.

(a) Show that (f_n) is a monotone increasing sequence.

(b) Use the $\epsilon - M$ definition of convergence to show that

$$\lim_{n\to\infty} |f_n - f_{n+1}| = 0.$$

(c) Is the sequence (f_n) Cauchy? Justify your answer using definitions and theorems from lecture.

Question 7 (11 marks)

Let $f : \mathbb{R} \to \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} 3x - 2 & \text{if } x \in \mathbb{Q} \\ x^3 & \text{if } x \notin \mathbb{Q}. \end{cases}$$

(a) Using the $\epsilon - \delta$ definition of continuity, show that f is not continuous at 0.

(b) Using an $\epsilon - \delta$ argument, show that f is differentiable at $x = 1$.

Question 8 (12 marks)

Let $f : [-1, 1] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} x+1 & \text{if } -1 \leq x < 0 \\ 0 & \text{if } x = 0 \\ x-1 & \text{if } 0 < x \leq 1 \end{cases}$$

Let $n \in \mathbb{N}^+$ and let P be the partition

$$P = \left\{-1, -\tfrac{n-1}{n}, \ldots, -\tfrac{2}{n}, -\tfrac{1}{n}, 0, \tfrac{1}{n}, \tfrac{2}{n}, \ldots, \tfrac{n-1}{n}, 1\right\}$$

(so that $x_i = -1 + i/n$ for each i).

(a) Draw a picture of illustrating how to compute the lower Riemann sum $L(f, P)$ when $n = 4$. Include the graph of f, the partition P of $[-1, 1]$, and the rectangles for each partition. The sum of the area of these rectangles should equal $L(f, P)$.

(b) Let $n \in \mathbb{N}^+$. Compute the lower and upper Riemann sums with respect to the partition

$$P = \left\{-1, -\tfrac{n-1}{n}, \ldots, -\tfrac{2}{n}, -\tfrac{1}{n}, 0, \tfrac{1}{n}, \tfrac{2}{n}, \ldots, \tfrac{n-1}{n}, 1\right\}$$

(so that $x_i = -1 + i/n$ for each i).

(c) Is f Riemann integrable on $[-1, 1]$? Explain.

Note. *For Questions 9 and 10, you may assume without $\epsilon - M$ proofs that limits behave the way you expect.*

Question 9 (13 marks)

(a) For each of the following infinite series, use an appropriate test or theorem to prove that the series converges or diverges.

(i) $\displaystyle\sum_{n=2}^{\infty} \frac{1}{n\sqrt{\log n}}$

(ii) $\displaystyle\sum_{n=1}^{\infty} 2^n e^{-n}$

(b) Find the radius of convergence and interval of convergence for the power series

$$\sum_{n=1}^{\infty} \frac{nx^n}{4^n(n+1)^2}$$

Question 10 (9 marks)

Consider the function $f : (-8, \infty) \to \mathbb{R}$ defined by

$$f(x) = (8 + x)^{-1/3}$$

for each $x > -8$.

(a) Find the Taylor series expansion of f at 0.

You do not need to justify your formula for the coefficients by a rigorous proof by induction.

(b) Show that the Taylor series of f converges to f on the interval $[-4, 4]$.

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End of Exam — Total Available Marks = 100

You must tick this box if you have used extra pages ☐